



SFTP Integration Document

Version 1.0

CENTRAL KYC RECORD REGISTRY (CKYCRR 2.0)

Version History

Version	Date	Details
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Table of Content

1.	Overview.....	3
2.	Prerequisites.....	4
3.	Configuration File Reference.....	4
4.	CLI Direct Execution	6
5.	Output Details.....	6
5.1	Exit Codes.....	6
5.2	Output files	7
6.	Standard Operating Procedure (SOP)	7
7.	Error Handling	8

1. Overview

In CKYCRR 2.0, Reporting Entities (REs) can exchange files with CKYCRR SFTP server using either of the following methods:

- **Web-Based** – A browser-based application that enables Reporting Entities (REs) to manually upload input Zip files and download processed response Zip files through a web-based user interface. The detailed procedure for the Web-Based approach is covered in the KYC Management document.
- **Programmatic Integration** – Enables the Reporting Entity's (RE's) internal application to invoke SFTPRunner.exe as a subprocess, thereby automating Zip file upload and download operations with the CKYCRR SFTP server without manual intervention.

Note: In both the case approval by second level RE user is required through Web based interface.
(Please refer user manual of KYC Management module for the same)

This document focuses on the Programmatic Integration approach using SFTPRunner.exe. SFTPRunner.exe is a self-contained utility which is platform agnostic designed to securely transfer files to and from the configured SFTP server using an embedded Java 21 runtime. Since all required runtime components are packaged within the executable, no separate Java installation or configuration is required on the client machine.

During execution, the utility performs the following operations:

- Reads the config.yaml configuration file from the specified location.
- Uses the embedded Java 21 runtime to execute all SFTP operations.
- Establishes a secure connection with the configured SFTP server.
- Uploads files from the configured local source directory to the designated SFTP location.
- Downloads processed output files from the configured SFTP directory to the configured local destination output folder, based on the selected execution mode.
- Tracks the status of each file and records whether the operation was completed successfully or failed.
- Generates a timestamped Executive Summary PDF containing a summary of the file transfer results.
- Creates a detailed timestamped log file capturing the complete execution details and any errors encountered.
- Returns a numeric exit code that enables the calling application to determine the execution outcome and take appropriate action.

Note: Invoke SFTPRunner.exe as a child process from the calling application. Once execution is complete, inspect the returned exit code (and standard output, if required) to determine whether the operation completed successfully or encountered any errors.

2. Prerequisites

Before invoking SFTPRunner.exe from the Reporting Entity's (RE's) internal application, ensure that the following prerequisites are met:

Requirement	Details
SFTPRunner.exe	Executable provided by CERSAI. Copy the executable to the designated deployment directory before invoking it from the RE's internal application.
config.yaml	Configuration file containing the SFTP connection details and utility settings. Place the file in the designated deployment directory or specify its location using the --config command-line parameter or the SFTP_CONFIG_PATH environment variable.
runtime\lib\sftp-utility.jar	Core SFTP engine required by the utility. Ensure the JAR file is available in the runtime\lib\ directory provided along with the executable.
runtime\jdk21\	Embedded Java 21 runtime provided by CERSAI. Copy the jdk21 folder to the deployment directory. No separate Java installation or configuration is required on the target machine.
Upload folder	Ensure the directory specified in config.yaml > upload.folder exists and contains the files to be uploaded to the CKYCRR SFTP server.
Download folder	Ensure the directory specified in config.yaml > download.path exists. Successfully processed Zip files downloaded from the CKYCRR SFTP server will be stored in this location.
Report folder	Ensure the directory specified in config.yaml > report.outputFolder exists. Execution reports generated by the utility will be stored in this location.
SFTP network access	Ensure the machine executing SFTPRunner.exe has network connectivity to the SFTP server configured in config.yaml.

3. Configuration File Reference

The config.yaml file contains the configuration parameters required for the execution of SFTPRunner.exe, including the SFTP connection details, file transfer locations, logging, and report generation settings.

Before executing the utility, open the config.yaml file using any text editor and update the configuration parameters as per the Reporting Entity's (RE's) deployment environment.

A sample config.yaml file is provided below for reference.

```

sftp:
  host: "https://sftp-uat.ckycindia.dev"
  username: "IAU000124"
  password: "Test@123"

upload:
  ficode: "IN0215"
  folder: "D:\\SFTP&FVU\\SFTP\\input"

download:
  path: "D:\\SFTP&FVU\\SFTP\\downloads"

report:
  outputFolder: "D:\\SFTP&FVU\\SFTP\\output"

```

Following keys will be used:

Key	M/O	Description
sftp.host	M	Specify the CKYCRR SFTP server URL corresponding to the deployment environment (UAT or Production), as provided by CERSAI.
sftp.username	M	Specify the SFTP username provided by CERSAI for the selected environment.
sftp.password	M	Specify the SFTP password associated with the configured username.
upload.ficode	M	Specify the Financial Institution (FI) Code assigned to the Reporting Entity (RE) by CERSAI.
upload.folder	M	Specify the local directory from which files will be uploaded to the CKYCRR SFTP server.
report.outputFolder	M	Specify the local directory where the execution summary report will be generated.
download.path	M	Specify the local directory where processed files (Success/Rejected) downloaded from the CKYCRR SFTP server will be stored.

4. CLI Direct Execution

Invoke SFTPRunner.exe as a child process from the Reporting Entity's (RE's) internal application. Wait for the execution to complete and inspect the returned exit code to determine whether the operation was successful or if any errors occurred.

Upon execution, the utility generates the following outputs:

- A timestamped Executive Summary PDF report in the configured report directory.
- A detailed timestamped log file containing the execution details and file transfer activities.
- A numeric exit code indicating the execution status.

```
:: Standard run - reads config.yaml from parent of exe folder
SFTPRunner.exe

:: Specify any config path
SFTPRunner.exe --config "C:\MyConfigs\production.yaml"

:: Using environment variable (set once per session)
set SFTP_CONFIG_PATH=C:\MyConfigs\production.yaml
SFTPRunner.exe
```

5. Output Details

5.1 Exit Codes

After SFTPRunner.exe completes execution, inspect the process exit code to determine the overall execution status.

The exit code indicates whether the utility completed successfully or encountered an error. The Reporting Entity's (RE's) internal application can use this value to determine the appropriate next action, such as marking the process as successful, retrying the operation, or recording the failure for further investigation.

Exit Code	Description	Recommended Action
0	File transfer completed successfully. All files were processed.	Mark the process as successful. The Executive Summary PDF report and processed output files can be found in the configured report.outputFolder and download.path directories, respectively.
1	Configuration error — missing or invalid config.yaml.	Verify the config.yaml file, correct the configuration, and execute the utility again. Review the generated log file for additional details.
3	Runtime exception — SFTP connection failed or JAR error.	Review the generated log file to identify the root cause, resolve the reported issue, and re-execute the utility.

5.2 Output files

The following files are generated after the successful execution of SFTPRunner.exe:

- Executive Summary PDF Report – A timestamped PDF report summarizing the upload/download execution results.
- Log File – A detailed timestamped log containing the execution details, file transfer status, and any errors encountered.
- Processed Output Files – Output response files downloaded from the CKYCRR SFTP server (Success/Rejected), available when the utility is executed in download mode.

File	Location	Description
Upload_Report_*.pdf	report.outputFolder	Executive Summary PDF report containing the details of the processed files.
upload_YYYYMMDD_HHMMSS.log	Configured log directory	Detailed run log — every step, JAR output line, and error. New file per run.
Processed Output Files	Download.path	Success and Rejected response files downloaded from the CKYCRR SFTP server.

6. Standard Operating Procedure (SOP)

Follow the steps below to execute **SFTPRunner.exe** from the Reporting Entity's (RE's) internal application:

Step 1: Ensure all prerequisite files (SFTPRunner.exe, config.yaml, lib folder, and jdk21 folder, if applicable) are available in the deployment directory.

Step 2: Update the configuration parameters in the config.yaml file, including the SFTP server details, local upload folder, local download folder, report output folder, and other required settings as per the deployment environment.

Step 3: Place the input files, generated after successful validation through the File Validation Utility (FVU), in the local directory configured under upload.folder.

Step 4: Invoke SFTPRunner.exe from the Reporting Entity's (RE's) internal application by specifying the upload execution mode. The utility uploads the files from the configured local upload.folder to the respective Financial Institution (FI) Code inbound directory on the CKYCRR SFTP server.

Step 5: File uploaded through SFTP mode will be available on the dashboard of RE Checker User. RE Checker user to login to CKYCRR 2.0 portal (www.ckycRegistry.co.in) using his/her login credentials and approve the uploaded file. After approval file uploaded will be processed by the system.

Step 6: Once the files are processed (Success/Rejected), invoke **SFTPRunner.exe** again from the Reporting Entity's (RE's) internal application by specifying the **download** execution mode. The utility downloads the processed response files from the CKYCRR SFTP server and stores them in the configured local **download.path** directory.

Step 7: Wait for the execution to complete and verify the returned exit code to determine whether the operation was successful or if any errors occurred.

Step 8: Review the generated output files:

- Executive Summary PDF Report in the configured report.outputFolder.
- Processed response files in the configured download.path directory (for download mode).
- Execution log file in the configured log directory for troubleshooting and audit purposes.

7. Error Handling

The following table describes the common errors that may occur during the execution of **SFTPRunner.exe**, along with their possible causes and recommended resolution steps.

Error code	Error cause	Resolution
Exit code 1 — "Config Validation Abort"	config.yaml is missing or contains invalid or incomplete configuration values.	Verify that the configuration file exists and all mandatory parameters are correctly configured.
Exit code 3 — Runtime Exception	Unable to establish a connection to the SFTP server, invalid credentials, or an internal processing error.	Verify the SFTP server details, user credentials, and review the execution log for the detailed error.
Missing sftp-utility.jar	The JAR was not copied into runtime\lib\.	Rebuilding the EXE from the project root — it copies runtime\ automatically.
Slow startup on every run	The EXE is unpacking its embedded JDK and JAR to a temporary folder before each run — this is expected behavior for a single-file build.	No action needed. If startup is unusually slow, confirm the exe was rebuilt with build.bat and that antivirus is not scanning the extraction on every launch.
No files uploaded, total = 0	The configured upload directory is empty or incorrectly configured.	Verify the value of upload.folder and ensure that files are available for upload.
PDF report not generated	report.outputFolder does not exist or is not writable.	Create the folder manually, or point outputFolder to a writable location.
EXE closes instantly with no output	The utility terminated unexpectedly during execution.	Review the generated execution log to identify the root cause.

Note: Review the execution log as the first step during troubleshooting. The log file contains detailed information about each execution, including connection status, file transfer activities, and any errors encountered.